

Risk-Based Audit Sample Selection: Michigan Medicaid Pharmacy Claims

Methodology Summary — Crystal Greer | Data Source: CMS State Drug Utilization Data (SDUD), Michigan, FY2024 (Q1–Q4)

OBJECTIVE

Identify pharmacy claims within Michigan's 2024 Medicaid drug utilization data that carry the highest risk of billing irregularities or non-compliance, and prioritize them for audit review — mirroring the core analyst function of researching and selecting audit samples based on risk indicators rather than reviewing claims at random.

DATA & SCOPE

The full Michigan SDUD file for FY2024 contains 131,357 claim-line records across both fee-for-service (FFSU) and managed-care (MCOU) utilization, spanning 2,988 distinct drug products. To focus the audit universe on the products with the greatest financial exposure — consistent with standard risk-based sampling practice — the analysis scoped to the 200 highest-volume products by total prescription count for the year, representing the drugs most likely to drive material dollar impact if billing errors exist.

RISK SCORING METHODOLOGY

For each of the 200 products in the audit universe, two independent risk indicators were calculated directly from the claims data:

- Cost Outlier Score — the product's average reimbursement per prescription, standardized as a z-score against the 200-product cohort. A high positive score flags a drug being reimbursed at an unusually high rate relative to its peers, a common signal of billing or pricing errors.
- Utilization Volatility Score — the swing between each product's highest and lowest quarter of prescription volume in 2024 (relative to its average), also standardized as a z-score. A high score flags erratic quarter-over-quarter utilization, which can indicate data entry errors, stockpiling, or emerging utilization patterns warranting review.

The two z-scores were summed into a single composite Risk Score, and all 200 products were ranked from highest to lowest risk. The top 30 highest-ranked products form the recommended audit sample.

KEY FINDINGS

- The 30 flagged products — 15% of the 200-product audit universe — account for approximately 70% of total dollars reimbursed within that universe, meaning the risk model concentrates audit effort on a small set of claims with outsized financial exposure.
- Cost-outlier drivers were concentrated among high-cost specialty drugs (e.g., long-acting injectable antipsychotics, GLP-1 weight-management medications), consistent with known categories of Medicaid drug spend growth.
- Utilization-volatility drivers included a mix of seasonal medications (e.g., antivirals with predictable seasonal demand swings) and rapidly-adopted newer therapies, both of which warrant different audit approaches — seasonal volatility may be expected and lower-priority, while volatility in a newly-utilized high-cost drug deserves closer review.

DELIVERABLE

The accompanying workbook (MI_Medicaid_Audit_Workbook.xlsx) contains four linked tabs: the full raw claims data, a formula-driven Risk Scoring model for the 200-product audit universe, the resulting High-Risk Audit Sample (top 30, ranked), and a Dashboard summarizing key metrics with a supporting chart. All calculations are live formulas referencing the raw data, so the workbook recalculates automatically if the underlying data changes.

LIMITATIONS

This is a claims-line risk-scoring exercise using publicly available aggregate CMS data, not a substitute for pharmacy-level or claim-level audit investigation, which would require access to individual claim documentation, prescriber and pharmacy identifiers, and contractual/regulatory reference data not present in the public SDUD file. It is intended to demonstrate a defensible, data-driven approach to audit sample prioritization.